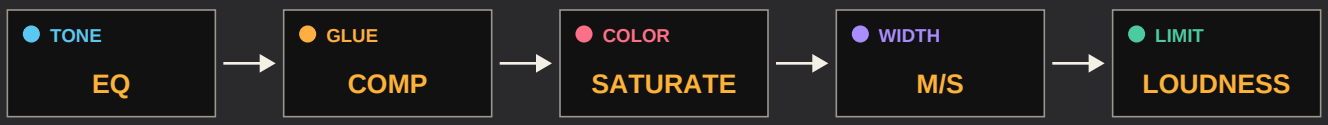
LACQUER

Mastering, from the start: getting your songs ready for release



Five stages, one knob, a loudness target

Mastering is the last step between a finished mix and a release. It is also the step that sounds most like a dark art, mostly because of the words around it: LUFS, true peak, limiting, glue. This guide explains each of them as it comes up. You need no background, just a mix you're happy with.

Lacquer is a mastering assistant: one plugin that does what a basic mastering chain does. You pick a **STYLE** (the character you want), a **TARGET** (where the song is going: streaming, a download store, CD or vinyl), press **LEARN** and play the song. It sets the levels so the master lands on the right loudness without a single peak going over. The **MASTER** knob then sets how much shaping goes on alongside. The **UNDER THE HOOD** strip shows what each of its five stages is doing, and you can switch any of them off to hear what it was adding. That is how you learn what each stage is for.

The first half of this guide is about mastering itself: what it is for, what loudness means and why it matters where the song is going. The second half gets Lacquer running in Live, tours the panel, walks through a session and ends with exporting files for distribution.

You'll need

- Ableton Live 11 or 12 (any edition that loads VST3 plug-ins).
- Lacquer built and installed. Building it copies **Lacquer.vst3** into ~/Library/Audio/Plug-Ins/VST3; a release download goes in the same folder.
- A finished mix, and headphones or speakers you know well.

> **The honest bit.** A mastering engineer brings trained ears, a treated room and years of comparing records. A plugin brings none of that. What Lacquer can do is get a good mix to a release-ready level safely, and show you what each step does so your ears learn what to listen for. For a release that really matters, it is still worth paying a person.

What mastering is

A mastering engineer gets a finished stereo mix and has three jobs:

- **Make it sound its best everywhere.** A mix made in one room, on one pair of speakers, gets small corrections so it holds up on earbuds, a car and a club system: a little less mud, a little more clarity, a low end that doesn't vanish on small speakers or boom on big ones.
- **Make it the right loudness for where it's going.** Streaming services, download stores, CDs and vinyl all want different things. More on this on the next page; it's the part that confuses everyone.
- **Make it safe.** No peaks over the top, no distortion when it's turned into an MP3 or AAC, and no bass that a vinyl cutting lathe can't handle.

On an album there is a fourth: make the songs sit together, so nobody reaches for the volume between tracks.

What it can't do

Mastering works on the whole mix at once, so it moves everything together. If the vocal is too quiet, the snare too loud or the bass muddy only on one note, mastering can't reach it without changing everything else as well. Fix it in the mix. A good rule: if the master needs more than about 2 dB of EQ anywhere, the mix needs another look.

The five stages

Most mastering chains run the same few tools in roughly the same order. Lacquer has five, and its UNDER THE HOOD strip shows them left to right in the order the sound goes through them:

Stage	The tool	What it's for
TONE	EQ (equaliser)	Small, broad tone corrections: some air on top, less mud in the low mids, a firmer low end. Plus a filter that removes rumble below 20 Hz you can't hear but that eats up level.
GLUE	Compressor	Turns loud moments down a little, usually a dB or two, so the mix moves as one and sounds more finished. The name comes from how a mix sounds "glued together" afterwards.
COLOR	Saturation	Gently rounds off peaks and adds harmonics, the way tape or analogue gear does. Reads as warmth and density.
WIDTH	Mid/side processing	A touch of extra width on the sides, and the deep bass kept in the middle so it hits the same on every system.
LIMIT	Limiter	Turns the whole thing up to the target loudness and stops any peak from going over the ceiling. This is where loudness comes from.

Every one of these is subtle on its own. That is normal: mastering is a stack of small moves. Switching a stage off and on while the song plays is the quickest way to hear what each one does.

Loudness, explained

Turn anything up and it sounds better for a moment: fuller, more exciting, more "finished". That trick is why, for twenty years, records got louder and louder, squashed until the drums stopped hitting. It was called the loudness war. Streaming ended it, and it is why loudness is measured carefully now.

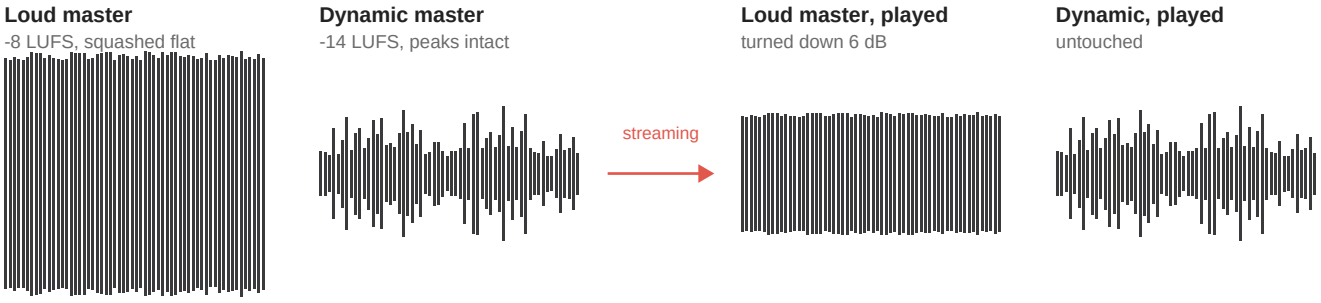
LUFS

LUFS (Loudness Units relative to Full Scale) measures how loud something **sounds**, not how high its peaks go. It weights the frequencies the way ears do, so a bass note counts for less than a vocal at the same level. Numbers are negative, and closer to zero is louder: -8 LUFS is very loud, -14 is typical for streaming, -23 is broadcast TV. One LU is one dB.

Reading	Over	Use it for
Integrated	The whole song	The number that matters: what a streaming service measures and what your target refers to.
Short-term	The last 3 seconds	How loud this section is: verse against chorus.
Momentary	The last 0.4 seconds	Moment to moment. It jumps around; that's fine.

Normalisation: why louder doesn't win any more

Spotify, Apple Music, YouTube, Tidal and the rest measure every song and turn it to the same loudness on playback. Spotify's default is -14 LUFS; Apple Music uses about -16. A master made at -8 LUFS gets turned down by 6 dB. All it keeps is what it lost to get that loud.



Both now play at the same loudness. The dynamic one still has its drum hits; the loud one gave them up for nothing.

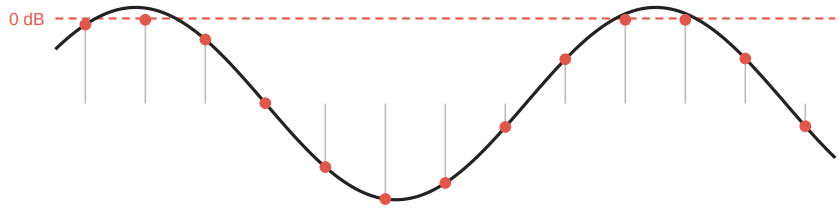
Spotify's own advice to artists is to aim for -14 LUFS integrated with true peaks below -1 dB, and below -2 dB if the master is louder than that. Services change their numbers now and then, so treat these as today's, not forever's.

Not everything is normalised. A download from Bandcamp, a WAV file or a CD plays at the level it was made at, next to whatever else is in the playlist, so masters for those usually sit louder. That is why Lacquer asks where the song is going.

True peak and the ceiling

Digital audio can't go above 0 dB full scale; anything over is clipped flat and distorts. But the samples are only points. The wave a player rebuilds between them can rise above the highest sample, and turning a file into MP3 or AAC makes those peaks grow further. **True peak** (dBTP) measures the rebuilt wave, not just the points. The

ceiling is the highest true peak allowed: -1 dBTP leaves room for the encoder.



Samples (dots) stop at 0 dB.
The wave a player rebuilds between them goes over: an inter-sample peak. A true-peak meter finds it; a sample-peak meter can't.

Lacquer's targets

TARGET	Loudness	Ceiling	Why
STREAMING	-14 LUFS	-1.0 dBTP	Matches how Spotify, YouTube, Tidal and Amazon play songs back. Nothing gets turned down, nothing is lost.
DOWNLOAD	-11 LUFS	-1.0 dBTP	Bandcamp and downloads play as made, so a bit louder holds its own in a playlist. Still MP3-safe.
CD	-10 LUFS	-0.3 dBTP	A CD plays exactly what's on it and is never re-encoded, so the ceiling can sit nearer 0.
VINYL	-13 LUFS	-1.0 dBTP	For a cutting engineer, who sets the final level. Moderate loudness, bass in mono below 150 Hz so the needle tracks.

One master for everything? Use STREAMING. It is the safe choice: it sounds right on every streaming service, and on a download it is just a little quieter than some.

Before you master

Get the mix ready

- **Leave some room at the top.** The loudest peaks of the mix should sit somewhere around -6 to -3 dB on Live's master meter. Lacquer adds the loudness; the mix doesn't need to be loud.
- **Take the limiter off your master channel.** If you've been mixing with a limiter or a maximiser on the master to hear it loud, switch it off before you export the mix. Two limiters in a row squash twice. (Glue compression you mixed into on purpose can stay.)
- **Export the mix as a WAV** at your project's sample rate, 24-bit or 32-bit float, with **Normalize** off. More on the export dialog on the last pages.

Two ways to set up

	How	Good for
A. On the mix	Put Lacquer last on the Master channel of the set you mixed in.	Learning, and quick passes. You can still go back and fix the mix with Lacquer running, which teaches you a lot.
B. A mastering set	Start a new Live set. Drag the exported mix WAV onto an audio track and put Lacquer on that track. Turn warping off for the clip.	Releases. The mix is fixed, the computer has nothing else to do, and the master channel stays clean. This is how engineers work.

Either way Lacquer must be **last**: nothing after it on that channel, and the Master fader at 0 dB. Anything after Lacquer can push peaks back over the ceiling.

Reference tracks

Pick two or three released songs in the same genre whose sound you like, and drag them onto another track in the set (not through Lacquer). They will be louder than your mix. Turn them down with a Utility until they sound about as loud as your master; solo between them and yours. This is the single most useful habit in mastering: it tells you whether your low end is too big, your top too dull, your master too squashed.

How to listen

- Pick a moderate volume and stick to it. Louder makes everything sound better, which tricks you.
- Take breaks. After twenty minutes ears adapt and stop noticing things.
- Check the finished master on other systems: earbuds, a phone speaker, a car. Mastering is about all of them, not just your studio.

Mono matters. Phones, many Bluetooth speakers and club systems play in mono or close to it. Press MONO in Lacquer now and then: if a part disappears or the mix goes hollow, something in the mix is too wide.

Set up

1 Let Live see Lacquer

Open Live › Settings (Cmd+comma) and choose the Plug-Ins tab. Turn on **Use VST3 Plug-In System Folders**, then press **Rescan**. In the browser, open Plug-Ins. Lacquer is under **Taddei**.

2 Put Lacquer last

Drag Lacquer to the end of the Master channel, or to the track holding your exported mix (see the previous page). Its panel opens; if it doesn't, click the wrench on the device. It starts on **Start Here**: GENTLE, STREAMING, MASTER at 50%.

3 Pick a STYLE and a TARGET

Choose the preset closest to your music from the list (Indie Rock, Hip-Hop, Ambient...), or pick a STYLE yourself. Then pick where the song is going. Not sure? GENTLE and STREAMING.

4 Press LEARN and play the song

Press **LEARN** (it blinks until you have) and play the song from the start. The display shows LEARNING and counts the seconds. Lacquer listens to the whole song and sets two levels: how hard the song hits its stages, and how hard it pushes the limiter to reach the target. When the song ends it stops by itself; you can also press LEARN again. Play all of it if you can: the chorus matters most, but the integrated loudness is measured over the whole song.

5 Play it again, and look at the numbers

Play the song from the start again. INTEGRATED counts up from the beginning (it resets each time you press play) and should finish within half a dB of the target: the line below it says **ON TARGET**. TRUE PEAK should stay at or under the ceiling. The short-term bar will swing either side of the red target mark; that is the song breathing, and it's fine.

6 Set MASTER by ear, and COMPARE

Turn **MASTER** up and down while the loudest part plays. Then press **COMPARE**: you hear your original mix at exactly the same loudness. Flick back and forth. If the master isn't clearly better, or you can hear it pumping, dulling the drums or getting harsh, turn MASTER down. Most songs land between 30% and 70%.

> **LEARN again** if you change STYLE or move MASTER a lot. Both change how loud the stages make the song, so the loudness drifts off target a little. Changing TARGET is fine: the level follows it, though a second LEARN gets it closest.

Find your way around



1	**Presets**	A STYLE, TARGET and MASTER for a kind of music. Loading one keeps what LEARN found. SAVE keeps your own.
2	**Style**	The character of the master. The text underneath says what the selected one does.
3	**Master**	How much of the style, on every stage at once. 0 = only the loudness is set.
4	**Target**	Where the song is going. The display shows the loudness and ceiling it aims for.
5	**Under the hood**	The five stages, showing what each is doing right now. Click one to switch it off. Meters show in and out.
6	**Learn**	Listens to the song and sets the levels for the target.
7	**Push**	Louder or quieter than the target, up to 6 dB.
8	**Loudness**	Integrated, the short-term bar with the target in red, momentary, true peak.
9	**Compare, mono**	The original at the same loudness; and the master in mono.
10	**Reset**	Restarts the integrated reading and true-peak hold. Pressing play does this too.

Hover over anything for a tip. TIPS turns them on and off.

Style, Master, Target

STYLE is a recipe for all five stages. MASTER decides how much of it to use: at 100% you get the whole recipe, at 50% every stage does half as much, and at 0% the stages rest and only the limiter works. The table shows each recipe at 100%.

	TONE	GLUE	COLOR	WIDTH	LIMIT
GENTLE	A little air and low end, a little less mud	1.5:1, slow	Barely any	+5%	Slow release, transparent
WARM	More low end, softer presence and top	2:1, slow	Tape-like, clearly there	None	Slow release
BRIGHT	Less mud, more presence, +2.5 dB air	2:1, medium	A little	+12%	Medium release
PUNCH	+2 dB low end, -2 dB mud, more presence	3:1, slow attack lets hits through	Some	+6%	Faster release
DENSE	Fuller low end, a little air	3:1, fast attack holds everything	The most	+10%	Fast release, dense

Choosing a style

- **Start with GENTLE** on anything. If it sounds right, stop: a light master is a good master.
- Does the mix sound **thin or cold**? Try WARM. **Dull or boxy**? BRIGHT. Do the **drums feel weak**? PUNCH. Should it be a **wall of sound**? DENSE.
- Compare the style against your reference tracks, not against your memory of them.

Setting MASTER

MASTER turns every stage up together, so it is the one knob to set by ear. The line under it tells you how much of which recipe is in use. Listen to the loudest section, turn it up until something gets worse (the drums lose their snap, the cymbals get harsh, the sound starts breathing in and out), then back it off a little. Use COMPARE to check you're still improving on the mix.

TARGET and PUSH

TARGET sets the loudness and ceiling (page 4). After LEARN, changing TARGET moves the level with it. **PUSH** sets the master louder or quieter than the target, by up to 6 dB. Turn it down for a quiet song on an album (a ballad shouldn't be as loud as the single). Turning it up squeezes harder through the limiter. Watch the LIMIT stage: past about 6 dB of gain reduction, most music starts to suffer.

Under the hood

Each block shows what its stage is doing right now. Click a block to switch the stage off and again to bring it back, and listen for the things below.

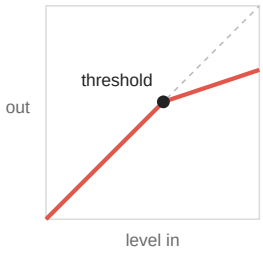
TONE — equalisation

First a high-pass filter removes everything below 20 Hz: rumble you can't hear that still uses up level. Then four broad bands: a low shelf at 90 Hz (weight), a dip or bump at 250 Hz (where mixes get muddy), presence at 3.2 kHz (vocals and attack) and an air shelf from 10 kHz (sparkle). Lacquer never moves any band more than 2.5 dB. The curve in the block is the EQ in use, from 20 Hz on the left to 20 kHz on the right; the line underneath names the two biggest moves.

Listen for: the low end tightening up, and the vocal stepping forward. If switching TONE off makes the song sound better, your mix wants a different STYLE.

GLUE — compression

A compressor turns the signal down when it gets loud. In mastering it works gently, a dB or two, so the mix holds together and loud moments don't jump out. Lacquer's ignores the deep bass when deciding (so a kick drum doesn't make the whole mix duck) and both sides move together so the stereo image stays put.



Above the threshold, each 3 dB louder in comes out only 1 dB louder (3:1).

Attack: how quickly it reacts. Slow lets the start of a drum hit through first: punch.

Release: how quickly it lets go. Too fast and you hear it breathing ("pumping").

Gain reduction: how much it is turning down right now. GLUE shows this number.

Listen for: with GLUE off, the mix gets a little looser and the loud hits a little more prominent. Gain reduction regularly past 3 dB is a lot for a master; turn MASTER down.

COLOR — saturation

Soft saturation bends the loudest parts of the wave a little, like tape or a valve stage. Peaks come out rounder and slightly lower, and new harmonics appear that the ear hears as warmth. The big number is how hard it's driven; the line underneath says how much it rounds the peaks. WARM and DENSE use the most.

Listen for: density and warmth, especially in the low mids. Too much sounds fuzzy or crunchy on cymbals and vocals.

WIDTH — mid/side

Any stereo signal can be split into a **mid** (what both speakers share) and a **side** (the difference between them). WIDTH turns the side up by a few percent for a wider image, and removes the side from the deep bass so it sits in the middle: solid on any system, and a must for vinyl, where wide bass makes the needle jump. The number is the width; the line underneath is the frequency below which the bass is mono.

Listen for: pads and reverbs spreading out a touch. Press MONO afterwards to check nothing disappears.

LIMIT — limiting

A limiter is a compressor that never lets anything through above its ceiling. Lacquer's pushes the song into it by the drive LEARN worked out (shown as push), so the song gets louder, and the limiter turns down just the peaks that would go over. It looks a couple of milliseconds ahead and checks between the samples, so even true peaks stay under. The big number is how much it's turning down right now.

Listen for: switching LIMIT off also removes the push, so it gets quieter. That is the point: almost all of the loudness comes from here. A few dB of gain reduction on the loudest hits is normal. If it is turning down a lot all the time, the master will sound flat and tiring.

The loudness section

LEARN

Press LEARN and play the song. While it listens, Lacquer measures the integrated loudness of the input and sets the **gain** into the stages, so GLUE and COLOR always see the same level however loud or quiet your mix was. At the same time it measures what comes out and adjusts the **drive** into the limiter until the output sits on the target. It stops after three seconds of silence once it has heard at least ten seconds, or when you press LEARN again. Less than five seconds isn't enough to judge, so it keeps the old settings.

What it learned is saved with your Live set, so the song comes back mastered. Presets don't change it, so you can try several on the same song. A new song needs a new LEARN.

The display

INTEGRATED	Loudness of everything since you pressed play (or RESET). Pauses don't count; quiet bits are weighed less, as the standard says. The line underneath compares it to the target.
Short-term bar	The last 3 seconds, from -36 to -4 LUFS. Green when within 1 LU of the target, yellow when over. The white tick is momentary; the red mark is the target.
MOMENTARY	The last 0.4 seconds.
TRUE PEAK	The highest peak since play, between samples included. Red if it ever went over the ceiling.

These follow ITU-R BS.1770 and EBU R128, the same measurement streaming services use, so Lacquer's INTEGRATED reading is what Spotify will see.

COMPARE, MONO, RESET

COMPARE plays your original mix in place of the master, delayed to line up and turned up or down to match its loudness (the display says by how much). Because louder always sounds better, this is the only fair comparison. If you prefer the original at the same loudness, the master is doing harm: lower MASTER, try another STYLE, or go back to the mix.

MONO folds the output to mono for checking. **RESET** restarts the integrated reading and the true-peak hold. Neither COMPARE nor MONO is saved with the set, so you can't export one by mistake.

> **Try this** Set MASTER to 0 and play the chorus: that is your mix at the target loudness and nothing else. Bring MASTER up slowly to 60% and listen to each stage come in. Then switch the stages off one at a time, from TONE to WIDTH, to hear what each was adding.

A session, step by step

1	Prepare	Mix peaking around -6 dB, no limiter on the mix. Load it into a mastering set, or put Lacquer last on the Master channel.
2	References	Two or three released songs on another track, turned down to match.
3	Style and target	Preset or STYLE; where it's going.
4	Learn	Press LEARN, play the whole song.
5	Master by ear	Set MASTER on the loudest section. COMPARE often. Switch stages off and on to hear each.
6	Check the numbers	Play from the top: INTEGRATED on target, TRUE PEAK under the ceiling.
7	Check elsewhere	MONO. Then export and listen on earbuds, a phone and a car.
8	Rest and revisit	Come back tomorrow with fresh ears before you send it anywhere.
9	Export	See the next page.

Common mistakes

- **Mastering louder than the target.** On streaming it just gets turned down. Use PUSH sparingly.
- **Judging by loudness.** Whatever is louder wins in a quick comparison. That is why COMPARE matches levels.
- **Fixing the mix in the master.** If the same complaint survives every STYLE, the fix is in the mix.
- **Mastering on one pair of headphones.** Check on at least two other systems.
- **Leaving a limiter on the mix bus** and then limiting again in Lacquer. Squashed twice.
- **Normalising on export.** It turns the whole file up until the highest peak hits 0 dB and undoes the ceiling.

An album or an EP

Master each song with the same TARGET, LEARN on each, then listen across the gaps between songs. Loudness targets measure averages, so a quiet ballad mastered to -14 LUFS can end up feeling louder than it should next to a rock song. Use PUSH on each song until the album feels even when played in order. On streaming the services normalise an album as a whole when it plays in order, so those differences survive.

Exporting for release

In Live, set the loop brace around the whole song with a second or two to spare at the end (so the last reverb tail isn't cut), then choose File › Export Audio/Video. The settings that matter:

Setting	Set it to	Why
Rendered Track	Master (or the track Lacquer is on)	The output with Lacquer on it.
Render Start / Length	The whole song, plus the tail	Nothing cut off.
Normalize	**Off**	Normalising turns the file up to 0 dB and undoes the ceiling.
Convert to Mono	Off	
Sample Rate	Your project's rate (usually 44100 or 48000)	Distributors take either; changing rate on export can add peaks.
File Type	WAV	What distributors want. FLAC is fine for your own archive.
Bit Depth	24 for distributors; 16 for CD	Ask your distributor; most take 16 or 24.
Dither	Triangular when exporting at 16-bit; none at 24-bit	Dither adds a whisper of noise that hides the grit of cutting bits off. Do it once, only at the end.

Per destination

Going to	TARGET	Export
Spotify, Apple Music, YouTube... (via a distributor)	STREAMING	WAV, 24-bit (or what your distributor asks), project rate, no dither at 24-bit.
Bandcamp	DOWNLOAD (or STREAMING if the same file goes to streaming too)	WAV or FLAC, 24-bit or 16-bit with dither. Bandcamp makes the MP3s itself.
CD	CD	WAV, 16-bit, 44100 Hz, Triangular dither. Leave gaps and track order to the CD software.
Vinyl	VINYL	WAV, 24-bit, project rate. Tell the cutting engineer it's a pre-master: they set the final level, and may ask for one with even less limiting.
SoundCloud, your site	STREAMING	A 24-bit WAV. They make their own MP3s or AACs.

Check the file. Drop the exported WAV onto a new track in an empty set, put Lacquer on it with every stage off and MASTER at 0, and play it through. The display measures it: INTEGRATED should match what you mastered to, and TRUE PEAK should stay under the ceiling. (Don't LEARN on it; just read.)

Good to know

LEARN said "too short"	It needs at least five seconds of sound. Press LEARN, play the song, and let it run.
INTEGRATED is a bit off target	You changed STYLE or MASTER after LEARN. LEARN again. Or you started playback mid-song: INTEGRATED measures from where you pressed play.
It went quiet when I switched off LIMIT	Expected: the loudness comes from the limiter. Switching it off removes the push too.
TRUE PEAK went red	Something after Lacquer pushed the level: the Master fader above 0 dB, or another plugin after it. Lacquer itself keeps peaks under the ceiling.
It sounds squashed or tiring	Lower MASTER, or PUSH. Look at LIMIT's gain reduction: if it's working hard all the time, the target is louder than this mix wants. Try STREAMING.
It sounds harsh	Try WARM, or lower MASTER. If it was harsh before Lacquer, the mix needs it fixed.
The bass sounds different in MONO	Some low end was out of phase in the mix. WIDTH puts the deepest bass in mono, but parts above its crossover are still stereo.
The master doesn't sound better in COMPARE	Trust it. Try less MASTER, another STYLE, or no change at all: a mix that needs nothing is a win.
Latency	Lacquer delays the sound by about 2 ms and tells Live, which lines everything up.
Presets	Your own are files in ~/Library/Application Support/Lacquer/Presets. The preset menu can show you the folder.

Glossary

dB	Decibel: a ratio of levels. +6 dB is about twice the signal; +10 dB sounds about twice as loud.
dBFS	Decibels below digital full scale. 0 dBFS is the top; everything is below it.
LUFS, LU	Loudness as heard (page 3). 1 LU = 1 dB.
True peak, dBTP	The peak of the wave rebuilt between the samples.
Ceiling	The highest true peak the limiter allows.
Headroom	The space between the loudest peak and 0 dBFS.
Dynamics	The difference between the quiet and loud parts. Compression and limiting reduce it.
Gain reduction	How much a compressor or limiter is turning the signal down, in dB.
Normalisation	A player turning every song to the same loudness. Also, on export, turning a file up so its peak hits 0 dB: leave that off.
Mid / side	The parts of a stereo signal both speakers share, and the difference between them.
Dither	A trace of noise added when cutting bit depth, so quiet sounds fade smoothly instead of crunching.